

1. Introduction

Objective:

- Select the predictor variables that result in a ‘best’ model within the scientific context of the problem
 - A basic plan to select a subset of variables for the model
 - A set of methods for assessing the adequacy of the model in terms of its individual variables and its overall fit

Variable Selection

- i. A selection process should begin with a careful univariate analysis of each variable.
 - a. Estimation of coefficients
 - b. Testing of coefficients for significance (Wald’s test)
 - c. Interpretation using odds ratio
- ii. Any variable whose univariate test has a p-value $< .25$ is a candidate for the multiple logistic regression model.
 - a. Sometime a collection of predictor variables, each of which is weakly associated with the response variable, can become an important predictor of response variable when taken together.
- iii. Stepwise variable selection
 - a. Forward selection
 - b. Backward elimination
- iv. Best subsets selection is a selection method that allows several models containing different number of predictor variables.
- v. Once a multiple logistic regression model is selected, importance of each variable included in the model should be verified.
 - a. Wald test for each predictor variable
 - b. Comparison of each estimated coefficient with coefficient from the model containing only those variables.
 - c. Variables that do not contribute to the model should be eliminated and a new model should be fit.
- vi. Once a model we think is adequate has been selected, we should look more closely to the variables

Stepwise Logistic regression

Forward Selection

- Step 1 – Let us say we have k -predictor variables
 - Fit the intercept only model
 - Followed by fitting each of the k -predictor variables one by one
 - Most important variable is the one with the smallest p-value
 - This variable $x_{j(\min)}^1$ is the one goes in the model first → minimum p-value
- Step 2 -Identify a p-value ($=.25$)
 - Fit the logistic model with $x_{j(\min)}^1$ and each of the remaining $k-1$ variables separately.
 - Select the minimum p-value from these $k-1$ models
 - If this minimum p-value is smaller than 0.25, add this variable in the model-
 $x_{i(\min)}^2$
- Step 3 – Continue the process until the p-value becomes greater than 0.25

Backward elimination

- Step 1 – Let us say we have k -predictor variables
 - Fit the full model with all k -predictors
 - Least important variable is the one with the largest p-value
 - This variable $x_{-j(\max)}^1$ is the one goes out from the model first max p-value
- Step 2 -Identify a p-value ($=.85$)
 - Fit the logistic model without $x_{j(\min)}^1$ and with the remaining $k-1$ variables.
 - Select the maximum p-value from these $k-1$ models
 - If this minimum p-value is larger than 0.25, remove this variable in the model-
 $x_{-i(\max)}^2$ max
- Step 3 – Continue the process until the p-values becomes smaller than ~~0.25~~ .85

Best Subset Models

- $AIC = 2(k+1) + \text{Deviance}$ → number of predictors
- $BIC = \log(n) * (k+1) + \text{Deviance}$